

Pharmaniaga Pyrazinamide Tablet

DESCRIPTION

A white to off white, round, flat bevelled edge tablet with a breakline on one side and plain on the other.

COMPOSITIONS

Each tablet contains Pyrazinamide 500 mg.

ACTIONS

Pyrazinamide exhibits tuberculostatic activity *in vitro* only at slightly acidic pH. The growth of tubercle bacilli within monocytes *in vitro* is completely inhibited by pyrazinamide at a concentration of 12.5 µg/ml.

Pyrazinamide is active only at an acid pH, and it is therefore active mainly on the tubercle bacilli located within the cell. It is these bacteria which are probably responsible for microbial persistence and thus for relapses after chemotherapy has stopped.

Pyrazinamide is readily absorbed from the gastrointestinal tract. Peak serum concentration occur about 2 hours after administration orally and have been reported to be about 33 µg per ml after 1.5 g and 59 µg per ml after 3 g. Pyrazinamide is approximately 10% bound to plasma proteins. The half life has been reported to be about 9 to 10 hours.

Pyrazinamide is widely distributed in body fluids and tissues and diffuses into the CSF. The CSF concentration is approximately equal to concurrent steady-state plasma concentrations in patients with inflamed meninges. It is metabolised primarily in the liver by hydrolysis to the major active metabolite pyrazinoic acid which is subsequently hydroxylated to the major excretory product 5-hydroxypyrazinoic acid. It is excreted through the kidney mainly by glomerular filtration. About 70% of a dose appears in the urine within 24 hours mainly as metabolites and 4% as unchanged pyrazinamide within 24 hours.

INDICATIONS

Indicated in combination with other antituberculars, in the treatment of tuberculosis after failure with the primary medications. It is effective only against mycobacteria.

CONTRAINDICATIONS

Pyrazinamide is contra-indicated in patients with:

- hypersensitivity to pyrazinamide or to any of the excipients

- hepatic disease
- hyperuricaemia and/or gouty arthritis
- acute porphyria

ADVERSE REACTIONS

General

Fever, porphyria and dysuria have rarely been reported. Gout (see Precautions).

Gastrointestinal (GI)

The principal adverse effect is a hepatic reaction. Hepatotoxicity appears to be dose related and may appear at any time during therapy. GI disturbances including nausea, vomiting and anorexia have also been reported. Hepatic reaction also varies from a symptomless abnormality of hepatic cell function, detectable only by laboratory tests, through a mild syndrome of fever, anorexia, malaise, liver tenderness, hepatomegaly and splenomegaly, to more serious reactions such as clinical jaundice, and rare cases of hepatic failure and death.

Hematologic and Lymphatic

Thrombocytopenia and sideroblastic anemia with erythroid hyperplasia, vacuolation of erythrocytes and increased serum iron concentration have occurred rarely with this drug. Adverse effects on blood clotting mechanisms have also been rarely reported.

Other

Mild arthralgia and myalgia have been reported frequently. Hypersensitivity reactions including rashes, urticaria, pruritis and aggravation of peptic ulcer have been reported. Fever, acne, flushing, photosensitivity, porphyria, dysuria and interstitial nephritis have been reported rarely.

WARNINGS AND PRECAUTIONS

Warnings:

Patients started on pyrazinamide should have baseline serum uric acid and liver function determinations. Those patients with preexisting liver disease or those at increased risk for drug related hepatitis (e.g., alcohol abusers) should be followed closely.

Pyrazinamide should be discontinued and not be resumed if signs of hepatocellular damage or hyperuricemia by an gouty arthritis appear.

Precautions:

General

Pyrazinamide inhibits renal excretion of urates, frequently resulting in hyperuricemia which is usually asymptomatic. If hyperuricemia is accompanied by acute gouty arthritis, pyrazinamide should be discontinued and not reinstated.

Pyrazinamide should be used with caution in patients with a history of diabetes mellitus, as management may be more difficult.

Primary resistance of M. tuberculosis to pyrazinamide is uncommon. In cases with known or suspected drug resistance, in-vitro susceptibility tests with recent cultures of M. tuberculosis against pyrazinamide and the usual primary drugs should be performed. There are few reliable in-vitro tests for pyrazinamide resistance. A reference laboratory capable of performing these studies must be employed.

Reduction in the size and/or frequency of dose is recommended for patients with renal insufficiency.

In general, dose selection for an elderly patients should be cautious, usually starting at the low end of the dosing range, reflecting the greater frequency of decreased hepatic or renal function, and of concomitant disease or other drug therapy.

Pyrazinamide should be used with caution in pregnant women.

Patients with rare hereditary problems of galactose intolerance, total lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Information for Patients

Patients should be instructed to notify physicians promptly and advised to discontinue treatment and seek immediate medical attention if they experience any of the following: fever, loss of appetite, malaise, nausea and vomiting, darkened urine, yellowish discoloration of the skin and eyes, pain or swelling of the joints.

Compliance with the full course of therapy must be emphasized, and the importance of not missing any doses must be stressed.

Laboratory Test

Side effects for pyrazinamide primarily involve the liver and vary from asymptomatic elevation of liver

function tests to serious clinical manifestations of hepatic disease. Baseline liver function studies [especially ALT (SGPT), AST (SGOT) determinations] and uric acid levels should be determined prior to therapy. Appropriate laboratory testing should be performed at periodic intervals and if any clinical signs of symptoms occur during therapy.

Drug/Laboratory Test Interactions

Pyrazinamide has been reported to interfere with ACETEST and KETOSTIX urine tests to produce a pink-brown color.

PREGNANCY AND LACTATION

Pregnancy

Pyrazinamide should only be used if the potential benefit justifies the risk to the foetus.

Lactation

Pyrazinamide has been found in small amounts in human breast milk. Therefore, it is advised that pyrazinamide be used with caution in nursing mothers taking into account the risk-benefit of this therapy.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

There are no data to suggest that Pyrazinamide affects the ability to drive or use machines.

Pyrazinamide may reduce the contraceptive effects of oestrogens and should be avoided 3 days before and after oral typhoid vaccination since it may inactivate the vaccine.

DOSAGE AND ADMINISTRATION

Recommended doses by mouth for adults range from 15 mg to 35 mg per kg body-weight to a maximum of 3 g daily given in 3 or 4 equally spaced doses. Doses of 20 mg per kg have been used for children when there is no satisfactory alternative.

OVERDOSAGE

Symptoms:

Extended effect of the adverse effects. Liver toxicity, hyperuricemia and abnormal liver function tests may occur with overdosage.

Treatment:

The stomach should be emptied by gastric lavage if necessary.

There is no specific antidote. General supportive measures should be employed. Liver function should be monitored closely, and a high-carbohydrate, low - fat diet employed. Care should be taken to avoid exposure of the patient to other hepatotoxic agents, including alcohol. Benzodiazepines may be given if there is evidence of central nervous system stimulation.

Probenecid may be given for hyperuricaemia.

INSTRUCTIONS FOR USE

Take the prescribed medicine at regular intervals preferably after meals.

STORAGE CONDITIONS

Store below 30°C.
Protect from light.

ROUTE OF ADMINISTRATION

Oral.

SHELF LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

PRESENTATION

In boxes of 100 tablets (10x10's) and 500 tablets (50x10's).

Date of revision: 29th July 2022

PRODUCT	REGISTRATION	HOLDER/
MANUFACTURER:		
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